# Akoizo Original Practice - 2027 Rotation Form 3 - Answer key

### Division C | Chemistry Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

### ANSWER KEY - Review only after submitting.

**1. (2 points) At constant temperature, a gas at 1.6 atm occupies 7 L. Compress it to 3.5 L. Find the final pressure.**

answer: 3.2 atm (2 points).

Numerical target 3.2; absolute tolerance 0.0064. Accepted units: , atm.

**2. (2 points) At constant pressure, a gas occupies 7 L at 360 K. Find its volume at 720 K.**

answer: 14 L (2 points).

Numerical target 14; absolute tolerance 0.028. Accepted units: , L.

**3. (2 points) A rigid vessel contains a gas at 1.6 atm and 360 K. Find its pressure at 180 K with no gas lost.**

answer: 0.8 atm (2 points).

Numerical target 0.8; absolute tolerance 0.0016. Accepted units: , atm.

**4. (2 points) Use PV=nRT with R=0.08206 L atm/(mol K). P=1.6 atm, V=7 L and T=360 K. Find the amount of gas.**

answer: 0.3791263844883148 mol (2 points).

Numerical target 0.37912638; absolute tolerance 0.00075825277. Accepted units: , mol.

**5. (2 points) A dry gas mixture has partial pressures 1.6 atm and 0.8 atm. Find total pressure.**

answer: 2.4000000000000004 atm (2 points).

Numerical target 2.4; absolute tolerance 0.0048. Accepted units: , atm.

**6. (2 points) A mixture has 5 mol oxygen and 15 mol nitrogen at total pressure 7 atm. Find oxygen partial pressure.**

answer: 1.75 atm (2 points).

Numerical target 1.75; absolute tolerance 0.0035. Accepted units: , atm.

**7. (2 points) Gas collected over water has total pressure 103 kPa. Water-vapor pressure is supplied as 6 kPa. Find dry-gas pressure.**

answer: 97 kPa (2 points).

Numerical target 97; absolute tolerance 0.194. Accepted units: , kPa.

**8. (2 points) Two ideal gases at the same temperature have molar masses 4 and 16 g/mol. What is the diffusion-rate ratio light/heavy? Use Graham's law.**

answer: 2 (2 points).

Numerical target 2; absolute tolerance 0.004. Accepted units: , .

**9. (2 points) For 2 H2 + O2 -> 2 H2O, 5 mol O2 reacts completely with excess H2. How many moles of H2 are consumed?**

answer: 10 mol (2 points).

Numerical target 10; absolute tolerance 0.02. Accepted units: , mol.

**10. (2 points) Explain why gas-law temperature ratios must use kelvin rather than degrees Celsius.**

answer: Kelvin is proportional to absolute thermodynamic temperature. Celsius has an arbitrary offset; its ratios do not represent molecular kinetic-energy ratios. (2 points).

**11. (2 points) [State/National] Determine the order with respect to A from the supplied initial-rate data.**

Synthetic initial-rate data (M/s): [A]=0.16 M, [B]=0.10 M gives rate 0.0128; [A]=0.32, [B]=0.10 gives 0.0512; [A]=0.16, [B]=0.20 gives 0.0256. Assume rate=k[A]^m[B]^n.

answer: 2 (2 points).

Numerical target 2; absolute tolerance 0.004. Accepted units: , .

**12. (2 points) [State/National] Determine the order with respect to B from the supplied initial-rate data.**

Synthetic initial-rate data (M/s): [A]=0.16 M, [B]=0.10 M gives rate 0.0128; [A]=0.32, [B]=0.10 gives 0.0512; [A]=0.16, [B]=0.20 gives 0.0256. Assume rate=k[A]^m[B]^n.

answer: 1 (2 points).

Numerical target 1; absolute tolerance 0.002. Accepted units: , .

**13. (2 points) [State/National] Calculate k from the initial-rate data. Give its value in M^-2 s^-1.**

Synthetic initial-rate data (M/s): [A]=0.16 M, [B]=0.10 M gives rate 0.0128; [A]=0.32, [B]=0.10 gives 0.0512; [A]=0.16, [B]=0.20 gives 0.0256. Assume rate=k[A]^m[B]^n.

answer: 5 M^-2 s^-1 (2 points).

Numerical target 5; absolute tolerance 0.01. Accepted units: , M^-2 s^-1.

**14. (2 points) [State/National] Predict the rate at [A]=0.48 M and [B]=0.10 M.**

Synthetic initial-rate data (M/s): [A]=0.16 M, [B]=0.10 M gives rate 0.0128; [A]=0.32, [B]=0.10 gives 0.0512; [A]=0.16, [B]=0.20 gives 0.0256. Assume rate=k[A]^m[B]^n.

answer: 0.11520000000000002 M/s (2 points).

Numerical target 0.1152; absolute tolerance 0.0002304. Accepted units: , M/s.

**15. (2 points) [State/National] For a first-order reaction, k=0.05330769 s^-1. Use ln(2)=0.693 to calculate the half-life.**

answer: 13 s (2 points).

Numerical target 13; absolute tolerance 0.026. Accepted units: , s.

**16. (2 points) [State/National] A first-order reactant starts at 1.04 M. Find concentration after three half-lives.**

answer: 0.13 M (2 points).

Numerical target 0.13; absolute tolerance 0.00026. Accepted units: , M.

**17. (2 points) [State/National] A zero-order reaction starts at 1.3 M and consumes reactant at 0.02 M/s. Find concentration after 10 s.**

answer: 1.1 M (2 points).

Numerical target 1.1; absolute tolerance 0.0022. Accepted units: , M.

**18. (2 points) A catalyst increases rate without changing reactant/product free energies. Explain the mechanism and whether the equilibrium constant changes.**

answer: It offers a lower-activation-energy pathway and speeds approach to equilibrium. At fixed temperature it does not change the equilibrium constant. (2 points).

**19. (2 points) Equal masses of the same solid react with excess acid under matched conditions. Why can smaller particles react faster?**

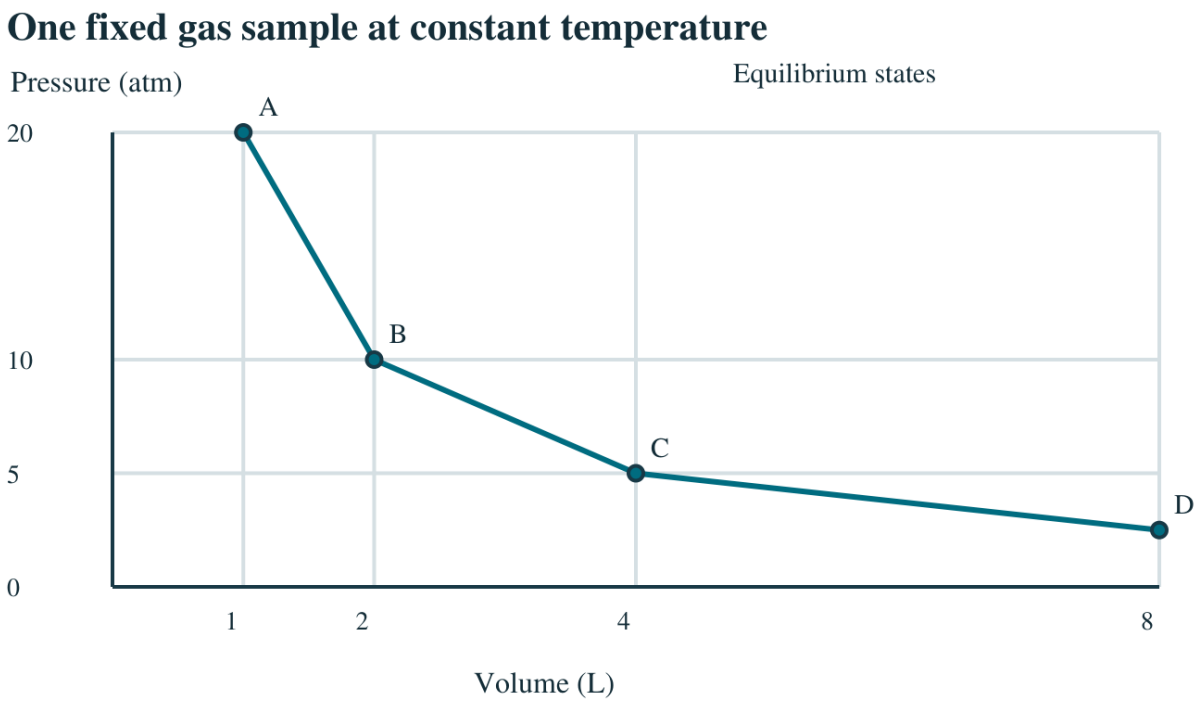
answer: Smaller particles expose more surface area, increasing contact and collision opportunities; mass alone does not set exposed area. (2 points).

**20. (2 points) [State/National] At 300 K a rate constant is k1; at 330 K it is 2k1. Using R=8.314 J/(mol K), ln(2)=0.693 and ln(k2/k1)=Ea/R(1/T1-1/T2), find Ea.**

answer: 19013.286599999985 J/mol (2 points).

Numerical target 19013.287; absolute tolerance 38.026573. Accepted units: , J/mol.

**21. (2 points) Use Figure F1. Read the pressure at state C.**




F1. Ideal gas isotherm (original synthetic teaching figure)

answer: 5 atm (2 points).

Numerical target 5; absolute tolerance 0.015. Accepted units: , atm.

**22. (2 points) Use Figure F1. Calculate PV at B.**

Refer to F1. Ideal gas isotherm (original synthetic teaching figure) shown with the first question in this figure group.

answer: 20 L atm (2 points).

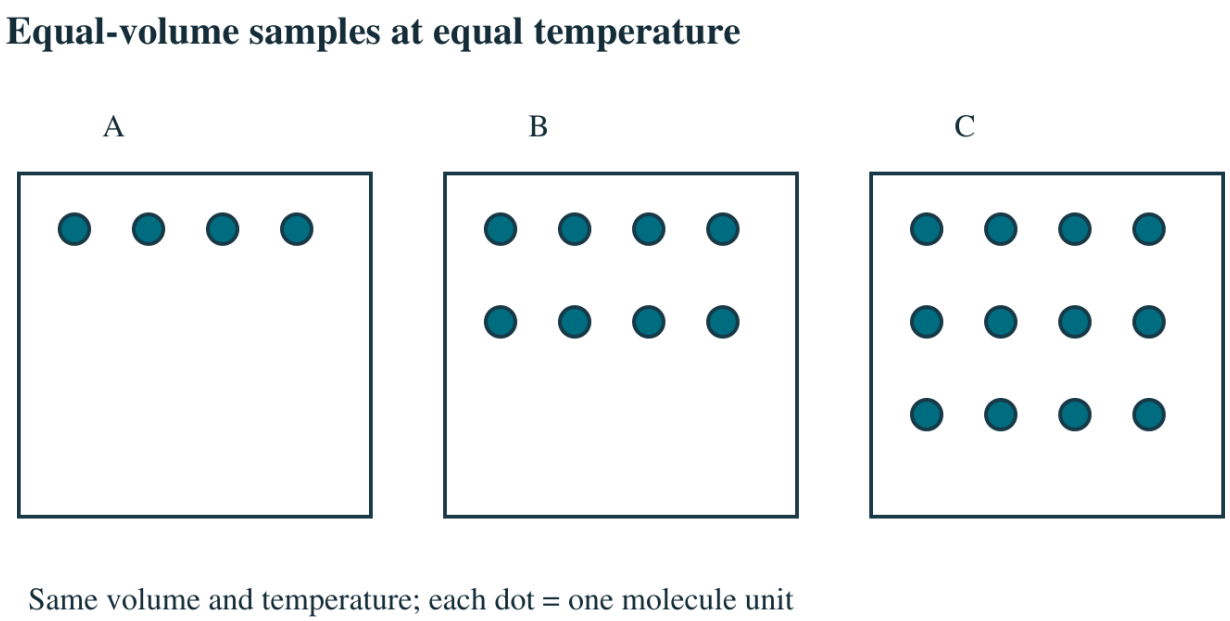
Numerical target 20; absolute tolerance 0.06. Accepted units: , L atm.

**23. (2 points) Use Figure F1. Use two plotted states to explain the volume-pressure relationship.**

Refer to F1. Ideal gas isotherm (original synthetic teaching figure) shown with the first question in this figure group.

answer: For example, volume doubles A to B while pressure halves; PV remains constant for this fixed amount at constant temperature. (2 points).

**24. (2 points) Use Figure F2. Find the pressure ratio B/A.**




F2. Ideal gas particle models (original synthetic teaching figure)

answer: 2 (2 points).

Numerical target 2; absolute tolerance 0.006. Accepted units: .

**25. (2 points) Use Figure F2. If A contains 0.10 mol, how many moles does C contain?**

Refer to F2. Ideal gas particle models (original synthetic teaching figure) shown with the first question in this figure group.

answer: 0.3 mol (2 points).

Numerical target 0.3; absolute tolerance 0.0009. Accepted units: , mol.

**26. (2 points) Use Figure F2. Would the dots justify a claim that C has faster molecules than A? Explain.**

Refer to F2. Ideal gas particle models (original synthetic teaching figure) shown with the first question in this figure group.

answer: No; equal temperature fixes the ideal-gas average translational kinetic energy, while this picture varies number density. Speed also depends on molecular mass. (2 points).

**27. (2 points) Use Figure F3. [State/National] Determine reaction order with respect to A from runs 1 and 2.**

## Matched-temperature initial-rate measurements

| Run | [A] (M) | [B] (M) | Rate (M/s) |
|---|---|---|---|
| 1 | 0.10 | 0.10 | 0.005 |
| 2 | 0.20 | 0.10 | 0.020 |
| 3 | 0.10 | 0.20 | 0.010 |

F3. Initial-rate experiment (original synthetic teaching figure)

answer: 2 (2 points).

Numerical target 2; absolute tolerance 0.006. Accepted units: .

**28. (2 points) Use Figure F3. [State/National] Determine reaction order with respect to B from runs 1 and 3.**

Refer to F3. Initial-rate experiment (original synthetic teaching figure) shown with the first question in this figure group.

answer: 1 (2 points).

Numerical target 1; absolute tolerance 0.003. Accepted units: .

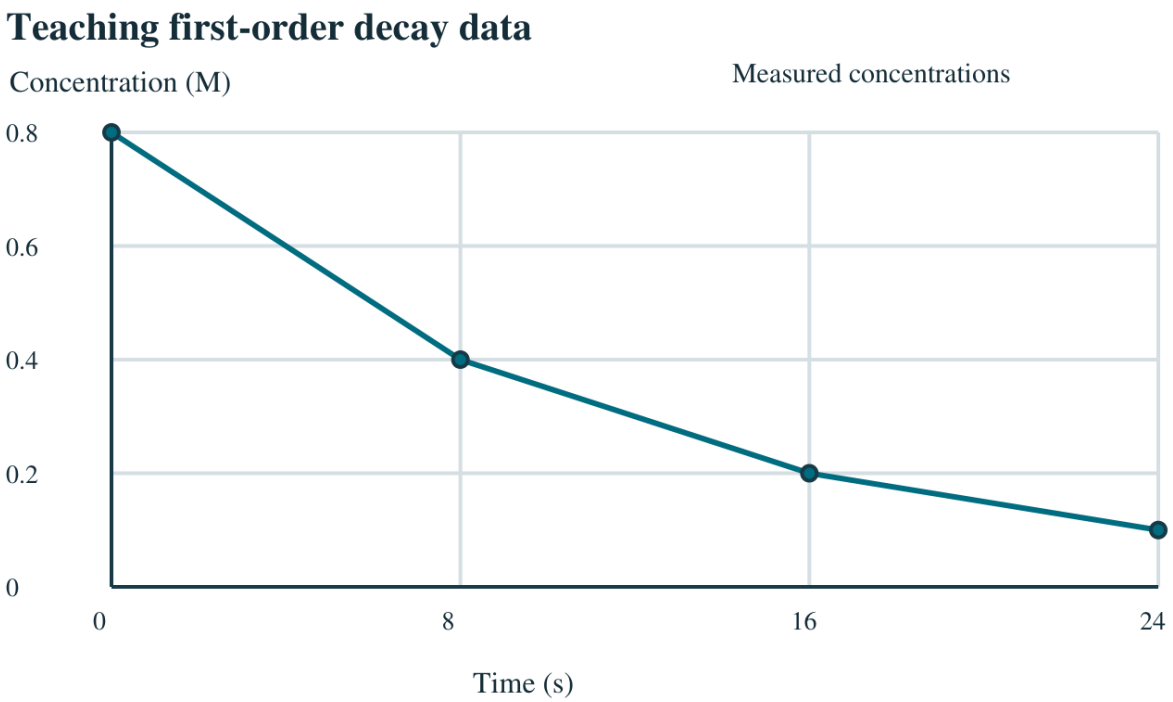
**29. (2 points) Use Figure F3. [State/National] Calculate k for rate=k[A]^2[B].**

Refer to F3. Initial-rate experiment (original synthetic teaching figure) shown with the first question in this figure group.

answer: 5 M^-2 s^-1 (2 points).

Numerical target 5; absolute tolerance 0.015. Accepted units: , M^-2 s^-1.

**30. (2 points) Use Figure F4. [State/National] Read the half-life.**




F4. Reactant decay curve (original synthetic teaching figure)

answer: 8 s (2 points).

Numerical target 8; absolute tolerance 0.024. Accepted units: , s.

**31. (2 points) Use Figure F4. [State/National] Use k=ln(2)/half-life with ln(2)=0.693 to calculate k.**

Refer to F4. Reactant decay curve (original synthetic teaching figure) shown with the first question in this figure group.

answer: 0.086625 s^-1 (2 points).

Numerical target 0.086625; absolute tolerance 0.000259875. Accepted units: , s^-1.

**32. (2 points) Use Figure F4. [State/National] Explain why the equal time intervals do not remove an equal concentration each time.**

Refer to F4. Reactant decay curve (original synthetic teaching figure) shown with the first question in this figure group.

answer: For first-order decay an equal fraction remains after each half-life; the absolute concentration decrement becomes smaller as concentration falls. (2 points).